



Full Size Acoustic Violin



Woody

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Summary

Notice: Please read this text carefully this is not a simple project, and it is possible to hurt yourself making this....

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Notice: Please read this text carefully this is not a simple project, and it is possible to hurt yourself making this. Please wear safety goggles while tightening and tuning for the first time, or do not hold the violin up by your face. A mouth guard wouldn't hurt either. While I personally haven't encountered problems, we all know 3D printed parts are not 100% reliable. Even though probably nothing will happen, if something does, you'll wish you had taken precaution. Violin strings can hold around 80 lbs of force. If the neck were to snap, it will hit pretty hard.

[Hear me play it](#)

[Read about its construction and material analysis here](#)

The link from the source/remix to original files has been broken. For the sake of preserving history, I'll [relink the original source](#).

I saw the redditors asking for the unsplit neck and scroll. It is added.

The scale between these files and their originals is 18500%

I've turned off the "work in progress" since everything I can think of is done, but I still might make minor changes. I'll have a longer video ~~in a few months~~eventually going into more depth about the violin. Maybe I'll make improvements to this a summer project.

Added a sound-post and bass-board. Used real strings. Bridge was thickened and shrunk to accommodate the neck height. Tail Piece needs thickening. Other pieces have been modified to accommodate stresses or proportions.

Used roughly 1.5 kg of plastic, although I don't know the total weight. Some parts like the sound-post and bridge may need scaling in one direction depending on how your violin turns out. I wish things could be more exact and repeatable, but this isn't a perfect violin. I don't recommend wood-fill, silk, or other irregular plastics. They usually have weaker mechanical properties, although they may still work.

Print Settings

Printer Brand:

Prusa

Printer:

i3 MK2S

Rafts:

No

Supports:

Yes

Resolution:

0.15 mm layer heights

Infill:

20% (Less is better. Less mass = Less inertia = less dampening = better sound)

Filament: Maker Geeks PLA

Notes:

I would guess that softer plastics like ABS would cause more sound dampening. A lighter and stiffer a material is, it is likely to sound better. Although less material is better, violins are under 50-80 lbs of stress from the strings, so it's a tough trade-off.

I print my PLA at 230 C.

Took about 3 days worth of print time, and 1 full day to assemble.

If you can, print the body-plates vertically. The current orientation is stronger, but also is worse for vibration.

Post-Printing

This is the method I originally used, although I do not believe it is the best. I put pieces into position with super-glue and clamps, then welded parts together with a soldering iron for stronger bonds. I would use extra lines of filament as needed. Better to have too much than have a snap. **Beware of toxic gases when burning super-glue, it will burn your eyes and lungs.** Use a respirator and sealed goggles. Tuning pegs were drilled to put the string through. I need to go see what size I used, and approximately where to drill. If you look up pictures online, you can get a pretty good rough guess.

I now believe it would be better to skip the super glue. There are better adhesives in the form of solvents. Things like Weld-On 3/4, or 3D Gloop will give cleaner and stronger bonds (Almost pure PLA). Melting with an iron may not be necessary anymore, but I haven't actually verified this.

The tail piece has a hole drilled through it and uses a copper wire to stay on the nipple.

Category: Music

This remix is based on



Full Size Acoustic Violin

by [Woody2552](#)

Model files



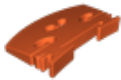
tailpiece.stl



sound_post_trimmed.stl



top2.stl



bridge.stl



tailend_pin_hole_vartiation.stl



bridgev2.stl



tailpiece_holes.stl



sound_post_tool.stl



top1.stl



bottom1.stl



neck_scroll_whole.stl



bottom2.stl



scrollandneck.stl



fingerboard.stl



side.stl



pegs.stl



sound_post_untrimmed.stl



chinrest.stl



bass_board.stl



tailend_pin.stl

[Find source .stl files on Thingiverse.com](#)

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